

Chromosome 17q12 Duplication Syndrome — Comprehensive Disease Characterization Report

Target Disease: Chromosome 17q12 Duplication Syndrome **MONDO ID:** MONDO:0013796
Category: Mendelian (recurrent genomic/copy-number disorder) **Report basis:** 11 confirmed findings, 37 primary papers reviewed over 5 iterations

Summary

Chromosome 17q12 duplication syndrome is a rare, autosomal-dominant **genomic (copy-number) disorder** caused by a recurrent ~1.4–1.5 Mb copy-number **gain** at chromosome band 17q12 (hg19 chr17:~34.8–36.2 Mb). The region is flanked by segmental duplications that predispose it to **non-allelic homologous recombination (NAHR)**, generating both this duplication and its reciprocal deletion. The duplicated interval spans roughly 15 OMIM genes, chief among them the dosage-sensitive transcription factor **HNF1B** (hepatocyte nuclear factor 1 beta), together with **LHX1** and **ACACA**. The disorder was first described alongside its reciprocal deletion in 2007 as "the first example of a recurrent genomic disorder associated with diabetes" ([PMID: 17924346](#)).

Clinically, the duplication produces a **highly variable, incompletely penetrant, predominantly neurodevelopmental/neuropsychiatric phenotype**. Reported features include a variable degree of intellectual disability/learning disability, delayed speech and language, delayed motor milestones, autism spectrum disorder, schizophrenia (odds ratio ≈ 4), epilepsy, ADHD, and — established more recently — tics and Tourette/persistent tic disorder (a genome-wide significant association). Because penetrance is incomplete and expressivity is wide, the duplication is frequently inherited from a mildly-affected or apparently unaffected parent. Congenital malformations are less consistent than in the reciprocal deletion but include prenatal duodenal atresia ("double bubble" sign), cardiac malformations (including tetralogy of Fallot), growth anomalies, and variable renal and esophageal anomalies.

Diagnosis rests on **chromosomal microarray analysis (CMA)** — the sub-microscopic (~1.4 Mb) CNV is invisible to standard karyotyping — with prenatal detection via CMA/CNV-seq on amniotic fluid, usually prompted by fetal ultrasound anomalies. Parental testing is essential given frequent inheritance. There is **no curative or disease-specific therapy**; management is supportive, symptom-directed, and multidisciplinary, with a notable pharmacovigilance caution against valproate given documented valproate-induced pancreatitis in this population and the syndrome's underlying pancreatic/metabolic susceptibility. Short-term prognosis appears generally favorable. The leading mechanistic hypothesis is **HNF1B gene-dosage imbalance**; a zebrafish ortholog (*vhnf1*) is dosage-sensitive in both loss- and gain-of-function directions and, when overexpressed, perturbs hindbrain patterning — supporting a gain-of-dosage mechanism relevant to the CNS phenotype.

Section 1 — Disease Information

Overview. 17q12 duplication syndrome is a recurrent contiguous-gene copy-number gain disorder. It is the reciprocal counterpart of the better-characterized 17q12 deletion syndrome (Renal Cysts and Diabetes syndrome, RCAD/MODY5). Whereas the deletion produces a relatively specific renal-and-diabetes phenotype, the duplication produces a **less specific, primarily neurodevelopmental/neuropsychiatric** phenotype with wide expressivity and incomplete penetrance ([PMID: 36548033](#); [PMID: 27409573](#)).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0013796
Cytogenetic locus	17q12 (hg19 chr17:~34.8–36.2 Mb)
Core gene	HNF1B (TCF2), OMIM 189907, HGNC:11630
Related deletion syndrome	17q12 deletion / RCAD (OMIM 137920)

Synonyms / alternative names: Chromosome 17q12 microduplication syndrome; 17q12 recurrent duplication; dup(17)(q12); reciprocal duplication of the 17q12 region.

Data provenance: The knowledge base for this disorder is drawn primarily from **aggregated disease-level resources** — case series, prenatal CMA cohorts, psychiatric CNV cohorts, and individual case reports — rather than a single EHR-derived individual-patient dataset. Evidence spans human clinical case series (majority), plus supporting model-organism (zebrafish, mouse) and in vitro data for mechanism.

Section 2 — Etiology

Primary cause (genetic). The disorder is caused by a **recurrent ~1.4–1.5 Mb duplication at 17q12 arising via NAHR** between flanking segmental duplications (Finding F001). Mefford et al. 2007 first described the reciprocal rearrangements and noted the duplication "appears to be enriched in samples from patients with epilepsy" (PMID: 17924346). The recurrent aberrations "encompass the genes, HNF1B, LHX1, and ACACA, among others" (PMID: 27409573).

Genetic risk factors. The duplication itself is the causal lesion; the dosage-sensitive driver is **HNF1B** (Finding F005). No independent susceptibility loci are established as necessary for expression, but **variable expressivity and incomplete penetrance** imply the action of modifier alleles and/or second genetic hits. Co-occurrence with independent pathogenic variants has been documented and can complicate the phenotype (e.g., co-inheritance with an *ARX* nonsense variant PMID: 32519823; with a *CCDC103* PCD mutation PMID: 26123568; a sibling pair with reciprocal 16p11.2 deletion and 17q12 duplication PMID: 34538867) — illustrating that a "second hit" may modulate outcome.

Environmental risk factors / protective factors. None established. As a Mendelian CNV disorder, there are no proven environmental causes, lifestyle risk factors, protective variants, or gene–environment interactions specific to disease occurrence. This is a genuine knowledge gap rather than a negative finding. (Advanced parental age is *not* associated with higher recurrent-CNV yield; in one prenatal cohort advanced maternal age carried a *lower* incidence, PMID: 38081620.)

Section 3 — Phenotypes

The duplication phenotype is dominated by neurodevelopmental and neuropsychiatric features, with variable congenital malformations (Findings F002, F004, F007, F008). Rasmussen et al. described "an extremely wide phenotypic spectrum, including a variable degree of learning disabilities, delayed language development, delayed motor milestones, and a broad range of psychiatric and neurological features" ([PMID: 27409573](#)).

Phenotype	Type	HPO suggestion	Onset	Frequency / notes
Intellectual disability / learning disability	cognitive	HP:0001249	childhood	Common; IQ range reported 52–99 (PMID: 30134084)
Delayed speech and language development	neurodevelopmental	HP:0000750	childhood	Common (PMID: 27409573)
Motor delay	neurodevelopmental	HP:0001270	infancy/ childhood	Common
Autism spectrum disorder	behavioral	HP:0000717	childhood	Recurrent (PMID: 22488896 ; PMID: 34538867)
Schizophrenia	psychiatric	HP:0100753	adolescence/ adult	OR $\approx$ 4.16 (P=0.018) (PMID: 24776740)
Epilepsy / seizures	neurological	HP:0001250	variable	Duplication "enriched in patients with epilepsy" (PMID: 17924346)
ADHD / behavioral abnormalities	behavioral	HP:0007018 / HP:0000708	childhood	Common
Tics / Tourette / persistent tic disorder	behavioral	HP:0100033	childhood	Genome-wide significant association (PMID: 40894066)
Structural brain abnormalities	CNS malformation	HP:0012443	congenital	Reported in shared del/dup features
	physical	HP:0001999	congenital	Reported

Phenotype	Type	HPO suggestion	Onset	Frequency / notes
Facial dysmorphism				
Joint laxity	connective tissue	HP:0001388	congenital	Reported
Renal anomalies	renal	HP:0000077	congenital	Variable (less consistent than deletion) (PMID: 21540130)
Duodenal atresia ("double bubble")	GI malformation	HP:0002247	prenatal	Prenatal duplication feature (PMID: 33678321)
Cardiac malformation (incl. tetralogy of Fallot)	cardiovascular	HP:0001636 / HP:0001631	prenatal	4/7 (57%) in one prenatal series (PMID: 39433644)
Esophageal atresia	GI malformation	HP:0002032	prenatal	Reported (PMID: 21540130)
Growth anomalies	growth	HP:0001507	prenatal/ childhood	Prenatal duplication feature (PMID: 33678321)

Characteristics. Severity is **variable** (mild to severe), progression is generally **stable** rather than degenerative (neurodevelopmental features are static, though psychiatric features such as schizophrenia and tics have their own developmental onset windows). Age of onset ranges from **congenital/prenatal** (malformations) to **childhood** (neurodevelopmental) to **adolescence/adulthood** (schizophrenia).

Quality of life impact. No disease-specific EQ-5D/SF-36/PROMIS data were identified. Impact is inferred to be driven by intellectual disability, behavioral/psychiatric burden, and — where present — surgical malformations. This is a knowledge gap.

Section 4 — Genetic / Molecular Information

Causal lesion and gene content (Findings F001, F005, F010). The canonical recurrent duplication maps to **hg19 chr17:~34.8–36.2 Mb** ([PMID: 40894066](#)); a representative case spanned chr17:34,460,444–36,243,365 (GRCh37) ([PMID: 39793343](#)). Reported CNV sizes range 1.42–1.91 Mb and "included 15 OMIM genes, such as HNF1B, LHX1, and ACACA" ([PMID: 39433644](#)).

Gene	HGNC / OMIM	Role
HNF1B (TCF2)	HGNC:11630 / OMIM 189907	Master transcription factor; kidney, pancreas, genitourinary development; leading dosage-sensitive driver
LHX1	HGNC:6593 / OMIM 601999	LIM-homeobox TF; genitourinary and neural development
ACACA	HGNC:84	Acetyl-CoA carboxylase alpha; lipid/fatty-acid metabolism
Other interval genes	—	AATF, DDX52, CCL3P, ZNHIT3, MYO19, PIGW, GGNBP2, DHRS11, MRM1, TADA2A, DUSP14, among others

Variant classification and type. The pathogenic event is a **structural variant (recurrent copy-number gain)**, not a point mutation — classified as **pathogenic** as a recurrent genomic disorder, though its clinical interpretation is complicated by incomplete penetrance and variable expressivity. Origin is **germline**. Functional consequence is **gene dosage increase (three copies)** of the interval genes; for HNF1B this is a **gain of dosage** (contrast with the reciprocal deletion's haploinsufficiency).

Allele frequency. As a recurrent NDD-CNV, the duplication is present at low frequency in the general/control population — a key reason its penetrance is now estimated lower than early case-control studies suggested ([PMID: 34817560](#); [PMID: 22130109](#)).

Modifier genes / epigenetics. No specific modifier genes or epigenetic marks are validated for the duplication. Variable expressivity implies modifiers exist; identifying them is a gap. Co-occurring independent variants (ARX, CCDC103, 16p11.2) act as phenotypic modifiers in reported families.

Chromosomal abnormality. By definition this is a **recurrent interstitial microduplication** (dup 17q12), NAHR-mediated, flanked by segmental duplications (Finding F001).

Section 5 — Environmental Information

Environmental factors, lifestyle factors, infectious agents: Not applicable. This is a Mendelian genomic disorder with no established toxic, occupational, radiation, dietary, behavioral, or infectious contributors to disease occurrence. The only environmental consideration identified is **iatrogenic** (drug exposure): valproic acid can precipitate pancreatitis in these patients (see Sections 9/12, Finding F009).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **NAHR between flanking segmental duplications at 17q12** → *results in* a recurrent ~1.4–1.5 Mb duplication (three copies of the interval) (Finding F001).
2. Extra copy of interval genes → *leads to* **increased gene dosage**, most consequentially of the dosage-sensitive transcription factor **HNF1B** (and LHX1, ACACA) (Finding F005).
3. Elevated HNF1B/LHX1 dosage during development → *perturbs* transcriptional programs that pattern multiple organ primordia. In zebrafish, *overexpression* of the HNF1B ortholog *vhnf1* "induces expansion of the val expression domain in the hindbrain," demonstrating that increased dosage disrupts **CNS (hindbrain) patterning** (Finding F011, [PMID: 11731484](#)).
4. Disrupted neurodevelopmental patterning → *results in* the **neurodevelopmental/neuropsychiatric phenotype** — intellectual disability, speech/motor delay, autism, epilepsy, ADHD, tics/Tourette, and increased schizophrenia risk (Findings F002, F007; [PMID: 27409573](#), [PMID: 24776740](#), [PMID: 40894066](#)).
5. **Branch — organogenesis:** dosage perturbation of HNF1B/LHX1 in kidney, pancreas, genitourinary, gut, and (inferred) cardiac primordia → *contributes to* variable **congenital malformations** — renal anomalies, duodenal atresia, cardiac defects (tetralogy of Fallot), esophageal atresia, growth anomalies (Findings F004, F008; [PMID: 33678321](#), [PMID: 21540130](#), [PMID: 39433644](#)).

6. **Modifier layer (inferred):** incomplete penetrance and variable expressivity → *modulated* by genetic background/second hits and stochastic developmental factors, explaining transmission from mildly/unaffected parents (Finding F003; [PMID: 30134084](#)).

Detail by category

- **Molecular pathways / cellular processes.** Core mechanism is **transcription-factor dosage dysregulation** during embryonic **regional specification of organ primordia** (GO:0009952 anterior/posterior pattern specification; GO:0007389 pattern specification process; GO:0048513 animal organ development). HNF1B is a master regulator of nephrogenesis and pancreatic/hepatic development. In zebrafish, "vhnf1 controls development of multiple organs through regulating regional specification of organ primordia" ([PMID: 11731484](#)).
- **Protein dysfunction.** No misfolding/aggregation; the defect is **quantitative overexpression** of otherwise normal proteins (gain of dosage). HNF1B is a POU-homeodomain transcription factor (UniProt P35680).
- **Metabolic changes.** ACACA (acetyl-CoA carboxylase) lies in the interval, linking the locus to fatty-acid/lipid metabolism; HNF1B dosage affects pancreatic development. These underlie the syndrome's **pancreatic/metabolic susceptibility**, clinically relevant to valproate-induced pancreatitis (Finding F009).
- **Immune involvement / tissue damage / oxidative stress:** not implicated — this is a developmental patterning disorder, not an inflammatory or degenerative one.
- **Molecular profiling.** No disease-specific transcriptomic/proteomic/metabolomic signature for the *duplication* was identified; mechanistic inference derives from the reciprocal deletion and model organisms. This is a gap.

GO term suggestions: GO:0009952, GO:0007389, GO:0048513, GO:0001822 (kidney development), GO:0031016 (pancreas development), GO:0021537 (telencephalon development).

CL term suggestions: CL:0000057 (fibroblast, for iPSC modeling), CL:0002518 (kidney epithelial cell), CL:0000164 (enteroendocrine/pancreatic lineage), CL:0000540 (neuron).

Section 7 — Anatomical Structures Affected

Level	Structure	UBERON / ontology	Involvement
Organ	Brain / CNS	UBERON:0000955	Primary — neurodevelopment, structural brain anomalies, hindbrain patterning
Organ	Kidney	UBERON:0002113	Variable — cystic/dysplastic/hypoplastic kidney, VUR
Organ	Pancreas	UBERON:0001264	Susceptible — pancreatic development; pancreatitis risk
Organ	Heart	UBERON:0000948	Malformation — tetralogy of Fallot, pulmonary artery anomalies
Organ	Duodenum / GI tract	UBERON:0002114	Duodenal atresia ("double bubble"), esophageal atresia
Organ	Esophagus	UBERON:0001043	Esophageal atresia
Body systems	Nervous, renal/urinary, cardiovascular, digestive, endocrine	—	Multi-system

Tissue/cell level. Predominantly **neural tissue** (neurons, CNS progenitors) and **epithelial tissues** of kidney, pancreas, and gut. **Subcellular:** HNF1B/LHX1 act in the **nucleus** (GO:0005634) as transcription factors. **Localization/lateralization:** malformations are variably unilateral or bilateral (e.g., unilateral multicystic kidney reported, [PMID: 42099279](#); bilateral hypoplastic kidneys reported, [PMID: 21540130](#)); CNS involvement is bilateral/diffuse.

Section 8 — Temporal Development

- **Onset.** Spans **prenatal/congenital** (malformations detectable on ultrasound), **infancy/childhood** (developmental delay, autism, epilepsy, ADHD, tics), and **adolescence/adulthood** (schizophrenia). Onset pattern is **insidious/chronic** for neurodevelopmental features.

- **Progression.** Neurodevelopmental features are largely **static/stable** (non-degenerative). Psychiatric features (schizophrenia, tics) follow their own developmental trajectories. Disease is **chronic/lifelong**.
 - **Course.** Not relapsing-remitting at the syndrome level; individual manifestations (epilepsy, tics, psychiatric episodes) may fluctuate.
 - **Critical periods.** The **embryonic/fetal window** of organ primordia specification is the mechanistically critical period (during which HNF1B dosage exerts its effect); postnatally, **early childhood** is the window for developmental/behavioral intervention.
 - **Prognosis note.** Despite incomplete penetrance, "short-term prognosis appears positive" (PMID: 39286125, Finding F009).
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Section 9 — Inheritance and Population

Epidemiology (Finding F003). The reciprocal 17q12 deletion has an estimated population prevalence of ~1:4,000; recurrent NDD-CNVs collectively occur in ~0.48% of newborns (~1 in 200 across 13 loci) (PMID: 32778765). The duplication is rarer and less precisely quantified but is on the order of 1 in several thousand. In prenatal CMA cohorts, 17q12 CNVs are recurrently detected among fetuses with ultrasound anomalies (PMID: 38081620).

Inheritance. - Pattern: Autosomal dominant. - **De novo vs inherited:** ~one-third de novo, ~two-thirds inherited. "Approximately a third of the newborn recurrent NDD CNVs (34%, N = 20/59) are de novo variants" (PMID: 32778765). The duplication "is often inherited from an apparently unaffected parent" (PMID: 30134084). - **Penetrance:** Incomplete. "17q12 copy number variants have variable presentations and incomplete penetrance, challenging prenatal counseling and management" (PMID: 39286125). - **Expressivity:** Highly variable, even within a single family (three-generation family, IQ 52–99; four affected children of a healthy mother, PMID: 30134084; PMID: 36548033). - **Anticipation / mosaicism / founder effects / consanguinity:** No genetic anticipation expected (not a repeat-expansion disorder). Germline mosaicism not specifically established for the duplication. No founder effect or consanguinity role — recurrence is driven by the NAHR-prone architecture, not by specific population haplotypes.

Population demographics. No strong ethnic predilection; recurrence is architecture-driven and thus pan-ethnic. **Sex ratio:** roughly balanced, though CNV pathogenicity may be sex-modulated for some neuropsychiatric outcomes ([PMID: 34817560](#)). No distinct geographic clustering.

Section 10 — Diagnostics

Cornerstone test — CMA (Finding F006). "17q12 deletions and duplications are two distinct, recurrent chromosomal aberrations usually diagnosed by chromosomal microarray analysis (CMA)" ([PMID: 27409573](#)); "Diagnosis is mostly established by chromosomal microarray" ([PMID: 38379631](#)).

Modality	Utility for 17q12 duplication
Chromosomal microarray (CMA)	First-line and definitive; detects the sub-microscopic ~1.4 Mb gain
CNV-seq	Equivalent prenatal alternative on amniotic fluid
Karyotyping	Insufficient — CNV is below cytogenetic resolution
FISH / MLPA / qPCR	Targeted confirmation of the specific CNV
Parental CMA	Essential — determines inheritance vs de novo, informs recurrence risk
WES/WGS	Not first-line for this CNV; useful when a co-occurring monogenic disorder is suspected (PMID: 40315683 , PMID: 40993696)

Prenatal pathway. Detection typically follows **fetal ultrasound anomalies** — renal (hyperechogenic kidneys), duodenal "double bubble," or cardiac findings — prompting amniocentesis and CMA/CNV-seq ([PMID: 33678321](#); [PMID: 39286125](#); [PMID: 38081620](#)).

Biomarkers / laboratory tests. No specific circulating biomarker. Metabolic surveillance (glucose, given HNF1B/pancreatic involvement) and renal function monitoring are advisable.

Imaging: renal ultrasound, echocardiography, and brain MRI as clinically indicated.

Differential diagnosis. Reciprocal 17q12 **deletion** (RCAD/MODY5 — renal cysts + diabetes, more specific phenotype); other recurrent NDD-CNVs (16p11.2, 15q11.2, 22q11.2, 1q21.1); isolated HNF1B point mutations; and co-occurring monogenic disorders that can mimic or compound the phenotype.

Screening. No population newborn screening. **Cascade testing** of relatives after a proband is diagnosed is appropriate given autosomal-dominant inheritance and frequent transmission from mildly-affected parents.

Section 11 — Outcome / Prognosis

- **Survival / mortality.** No excess early mortality is established for the isolated duplication; life-limiting risk derives from severe congenital malformations (major cardiac defects, complex GI atresia) when present. No disease-specific survival curves exist.
- **Morbidity / function.** Chief morbidity is **neurodevelopmental and psychiatric** — intellectual disability, autism, ADHD, epilepsy, tics, and elevated schizophrenia risk — producing variable long-term functional impairment. Congenital malformations contribute surgical morbidity.
- **Disease course.** Chronic and lifelong but generally **non-progressive**; "short-term prognosis appears positive" ([PMID: 39286125](#), Finding F009).
- **Prognostic factors.** Severity is unpredictable due to incomplete penetrance/variable expressivity; presence and severity of congenital malformations and degree of intellectual disability are the main determinants. No validated prognostic biomarkers.

Section 12 — Treatment

No curative or disease-specific therapy exists. Management is **supportive, symptom-directed, and multidisciplinary** (Finding F009): "Treatment involves a multidisciplinary approach" ([PMID: 38379631](#)).

Manifestation	Intervention	NCIT suggestion
Epilepsy	Antiseizure medication (avoid valproate where possible)	NCIT:C264 (Anticonvulsant)
Developmental delay / ID	Early developmental intervention, special education, speech & occupational therapy	NCIT:C15318 (Rehabilitation Therapy)
Autism / ADHD / behavioral	Behavioral therapy, psychiatric management	NCIT:C15328 (Behavioral Therapy)
Psychiatric (schizophrenia)	Standard psychiatric pharmacotherapy	NCIT:C265 (Antipsychotic Agent)
Renal/metabolic	Nephrology surveillance, glucose monitoring	NCIT:C15311 (Monitoring)
Duodenal/esophageal atresia	Surgical repair	NCIT:C15329 (Surgical Procedure)
Cardiac defects (e.g., TOF)	Cardiac surgical repair	NCIT:C15329

Pharmacovigilance caution (Finding F009). Valproic acid should be used cautiously. A 14-year-old with 17q12 duplication, focal epilepsy, ASD, and ADHD developed valproate-induced acute pancreatitis (lipase 1,572 U/L); "Discontinuation of the VPA led to rapid clinical improvement and normalization of lab values" (probable ADR by Naranjo scale) ([PMID: 41700275](#)). This is mechanistically plausible given the syndrome's pancreatic/metabolic susceptibility (HNF1B/ACACA in the interval).

Pharmacogenomics, gene/cell/RNA/targeted/immuno-therapies: Not applicable / none available. No advanced or experimental disease-modifying therapies or registered clinical trials specific to 17q12 duplication were identified. **Personalized medicine** currently amounts to genotype-informed surveillance (renal, metabolic) and drug selection (valproate avoidance).

Section 13 — Prevention

- **Primary prevention.** Not possible — the causal lesion is a germline CNV. The only actionable primary-prevention avenue is **reproductive**: prenatal diagnosis (CMA/CNV-seq) and **preimplantation genetic testing (PGT)** for carrier parents.
 - **Secondary prevention.** Early detection via prenatal ultrasound + CMA, and **cascade genetic testing** of relatives, enabling early developmental/surveillance intervention.
 - **Tertiary prevention.** Surveillance and management of complications — renal function monitoring, glucose/metabolic monitoring, timely surgical repair, and **avoidance of valproate** to prevent iatrogenic pancreatitis.
 - **Genetic counseling (central).** Given autosomal-dominant inheritance, incomplete penetrance, variable expressivity, and frequent transmission from mildly/unaffected parents, counseling must convey that (a) a carrier parent has a 50% transmission risk, (b) an inheriting child's phenotype is unpredictable, and (c) parental testing is essential ([PMID: 30134084](#); [PMID: 39286125](#)).
 - **Immunization / public health / environmental interventions:** Not applicable.
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Section 14 — Other Species / Natural Disease

- **Taxonomy / natural disease.** No naturally occurring 17q12-duplication syndrome is described in companion animals or wildlife (OMIA: none identified). The disorder is human-specific in its recurrent-CNV form because it depends on the human-specific segmental-duplication architecture flanking 17q12.
 - **Orthologous genes.** HNF1B ortholog in mouse (*Hnf1b*, NCBI Gene 21410) and zebrafish (*vhnf1/hnf1ba*). LHX1 and ACACA are conserved across vertebrates.
 - **Comparative biology / conservation.** HNF1B's developmental role is **evolutionarily conserved**: the zebrafish ortholog regulates gut, pronephros, and hindbrain specification ([PMID: 11731484](#)), indicating conserved dosage sensitivity of the mechanism.
 - **Zoonotic potential / cross-species transmission:** Not applicable (non-infectious genetic disorder).
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Section 15 — Model Organisms

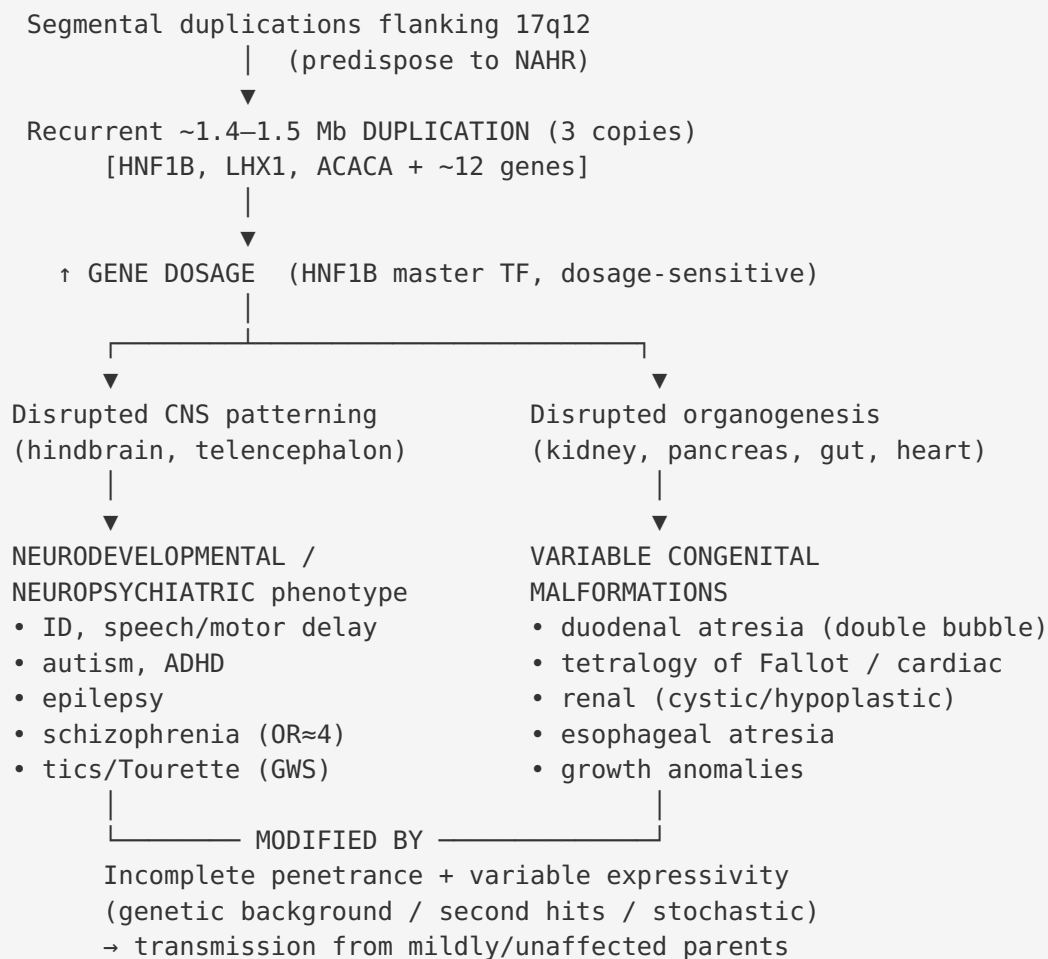
Zebrafish (Finding F011). The strongest gain-of-dosage evidence comes from zebrafish *vhnf1* (the MODY5/GCKD ortholog). Mutants show "formation of kidney cysts, underdevelopment of the pancreas and the liver, and reduction in size of the otic vesicles," and critically, "overexpression of *vhnf1* induces expansion of the val expression domain in the hindbrain" — demonstrating dosage sensitivity in **both** loss- and gain-of-function directions and directly modeling how *increased* HNF1B dosage perturbs CNS patterning (PMID: 11731484). This is the most disease-relevant model for the **duplication** specifically.

Mouse. *Hnf1b* (MGI; NCBI Gene 21410): complete null is early-embryonic lethal; conditional/heterozygous models produce renal cystic dysplasia and pancreatic hypoplasia — modeling **haploinsufficiency (the deletion)** more than the duplication. Ciliary-gene knockout mice display an "adipopancreatosis" phenotype linking related pathways to exocrine pancreatic disease (PMID: 42613169).

Model gaps. No published mouse model of the full multigene 17q12 duplication was identified. iPSC/organoid models of patient-derived duplications would be a high-value addition. **Model databases:** MGI (*Hnf1b*), ZFIN (*vhnf1/hnf1ba*), Alliance of Genome Resources.

Model	Type	Recapitulation	Limitation
Zebrafish <i>vhnf1</i> overexpression	gain-of-function	Hindbrain patterning perturbation (CNS relevance)	Single-gene, not full CNV
Mouse <i>Hnf1b</i> het/conditional KO	loss-of-function	Renal cystic dysplasia, pancreatic hypoplasia	Models the deletion, not duplication
Patient iPSC/organoids	in vitro	(proposed)	Not yet reported for the duplication

Mechanistic Model / Interpretation



The unifying principle is **transcription-factor gene-dosage imbalance**. The reciprocal deletion (HNF1B haploinsufficiency → RCAD/MODY5, a renal-and-diabetes phenotype) and the duplication (HNF1B overexpression → predominantly neurodevelopmental phenotype) are two ends of one dosage spectrum. The zebrafish gain-of-function data are pivotal because they show that *too much* HNF1B-ortholog activity, not just too little, disrupts development — grounding the duplication's CNS phenotype in a demonstrated (not merely inferred) mechanism.

Evidence Base

PMID	Contribution	Supports finding
17924346	First description of reciprocal 17q12 rearrangements; duplication enriched in epilepsy; "first recurrent genomic disorder associated with diabetes"	F001
27409573	Danish cohort (38 patients); defines wide dup phenotype; genes HNF1B/LHX1/ACACA; CMA as diagnostic	F001, F002, F006
24776740	Swedish schizophrenia CNV study; 17q12 dup OR=4.16, P=0.018	F002
30134084	Multi-generation family; variable penetrance; inheritance from unaffected parent; IQ 52–99	F002, F003
32778765	12,252-trio newborn study; ~34% de novo; prevalence estimates	F003
39286125	Prenatal CMA cohort; incomplete penetrance; favorable short-term prognosis	F003, F009
33678321	Prenatal duplication features: double bubble, cardiac, growth anomalies	F004, F008
21540130	Renal malformations & esophageal atresia can occur with the duplication	F004
24487052	HNF1B as dosage-sensitive driver (kidney/pancreas/GU); RCAD	F005
38379631	Neuropsychiatric case report; CMA diagnosis; multidisciplinary treatment	F006, F009
40894066 / 42749776	Genome-wide significant 17q12 dup association with TS/persistent tic disorder	F007
39433644	Prenatal series; cardiovascular anomalies 4/7 incl. TOF; 15 OMIM genes	F008, F010
41700275	Valproate-induced pancreatitis in a 17q12 dup patient	F009
39793343	Precise GRCh37 coordinates of a 17q12 duplication	F010
11731484	Zebrafish <i>vhnf1</i> : dosage sensitivity; overexpression perturbs hindbrain	F011

Evidence-type distribution: Predominantly **human clinical** (case series, prenatal cohorts, psychiatric CNV cohorts, case reports), with supporting **model-organism** (zebrafish gain-of-function; mouse *Hnf1b*) mechanistic data. No disease-specific in vitro/omics profiling of the duplication was identified.

Limitations and Knowledge Gaps

1. **Penetrance and expressivity are quantitatively imprecise.** Population-based estimates are confounded by higher-than-expected control frequencies; per-phenotype penetrance figures for the duplication remain uncertain.
 2. **Frequency data are largely borrowed from the deletion** or from pooled recurrent-CNV cohorts; a duplication-specific prevalence is not firmly established.
 3. **No duplication-specific molecular profiling** (transcriptomics/proteomics/metabolomics) exists; mechanism is inferred from the reciprocal deletion, single genes, and model organisms.
 4. **No mouse model of the full multigene duplication;** the strongest gain-of-dosage evidence is from a single-gene zebrafish overexpression experiment.
 5. **Contributions of non-HNF1B genes** (LHX1, ACACA, others) to the duplication phenotype are poorly delineated.
 6. **No validated modifiers** explain the wide intrafamilial variability; second-hit contributions are documented anecdotally (ARX, CCDC103, 16p11.2) but not systematically.
 7. **No quality-of-life, disability, or long-term outcome instruments** have been applied to this population.
 8. **No disease-modifying therapy or clinical trials** — treatment is entirely symptomatic.
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Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC and cerebral/kidney organoid models** of the recurrent duplication to obtain duplication-specific transcriptomic/proteomic signatures and test HNF1B-dosage causality directly in human cells.

2. **Engineer a mouse (or zebrafish) model carrying the syntenic multigene duplication** to compare against single-gene *Hnf1b* overexpression and dissect gene-by-gene dosage contributions (LHX1, ACACA).
3. **Assemble a large, prospectively phenotyped 17q12-duplication registry** with parental testing to derive robust, per-phenotype penetrance and expressivity estimates and formal QoL/functional outcomes.
4. **Genetic-modifier and second-hit study**: WGS across variably affected carriers within families to identify modifier alleles explaining variable expressivity.
5. **Pharmacovigilance analysis** (FAERS + registry) to quantify valproate-associated pancreatitis risk in HNF1B-dosage disorders and formalize a valproate-avoidance recommendation.
6. **Systematic cardiac and GI malformation surveillance protocol** for prenatally diagnosed cases, given newly recognized tetralogy-of-Fallot and duodenal-atresia associations.
7. **Cross-locus comparison** of neuropsychiatric trajectories (schizophrenia, tics, ASD) between 17q12 duplication and other recurrent NDD-CNVs to refine genotype-specific counseling.

Report compiled from 11 confirmed findings and 37 primary papers. Evidence is predominantly human clinical, supported by zebrafish and mouse model-organism data for mechanism. Where information was not available or not applicable (environmental etiology, infectious agents, veterinary natural disease, advanced/experimental therapeutics), this is stated explicitly.
